Regional Neurogenetics Laboratory

FICTITIOUS EXAMPLE

Example Hospital, Example Road, Exampletown

Neurogenetics Report - Repeat Expansion Testing

Patient	Robert BLACKBURN
Date of birth	14-Jan-1965
NHS number	973 738 3370
Report number	NG24-01873
Report date	14-May-2024
Specimen	Peripheral blood (EDTA), received 22-Apr-2024

Clinical question

Progressive cerebellar ataxia from age 55, no family history. Testing for the common repeat expansion ataxias.

Results

All alleles tested are within the normal range.

Gene (disorder)	Repeat	Alleles	Normal range
ATXN1 (SCA1)	CAG	29, 30	35 or fewer
ATXN2 (SCA2)	CAG	22, 22	31 or fewer
ATXN3 (SCA3)	CAG	23, 27	44 or fewer
CACNA1A (SCA6)	CAG	11, 13	18 or fewer
ATXN7 (SCA7)	CAG	10, 12	27 or fewer
TBP (SCA17)	CAG/CAA	36, 38	40 or fewer
ATN1 (DRPLA)	CAG	15, 17	35 or fewer
FXN (Friedreich ataxia)	GAA	9, 17	33 or fewer

Interpretation

These results exclude spinocerebellar ataxia types 1, 2, 3, 6, 7 and 17, DRPLA and Friedreich ataxia as the cause of this patient's ataxia.

Method and limitations

Fluorescent PCR and fragment analysis, with repeat-primed PCR for FXN. Other repeat expansion loci, including RFC1 (CANVAS), were not tested. Normal ranges in this example are illustrative.

Authorised by

Dr A Example, Consultant Clinical Scientist